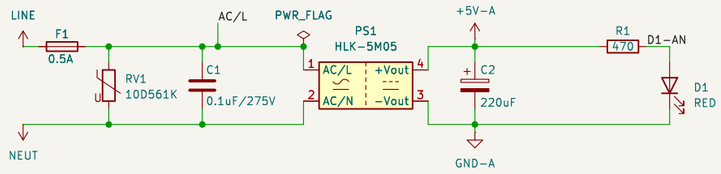
POWER SUPPLY 220VAC–5VDC

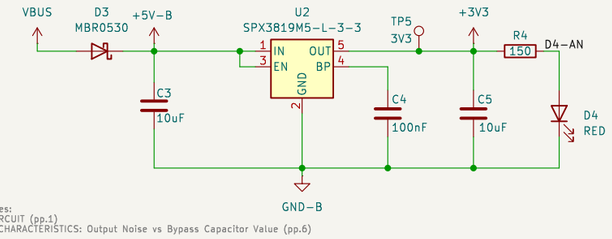


Reference:
HLK–5M05 Datasheet

Design reference on title:
8. Typical application circuit (pp.7)

PCB–A

LDO 5V–3V3

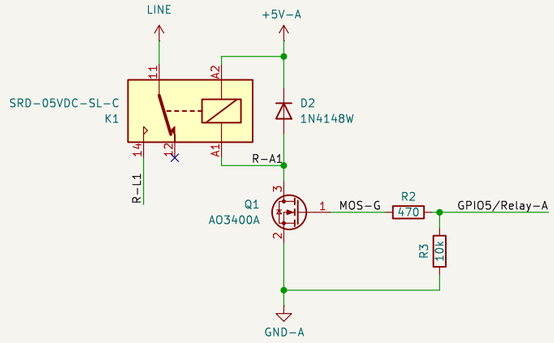


Reference:
SPX3819 Datasheet

Design reference on titles:
TYPICAL APPLICATION CIRCUIT (pp.1)
TYPICAL PERFORMANCE CHARACTERISTICS: Output Noise vs Bypass Capacitor Value (pp.6)

PCB–B

RELAY



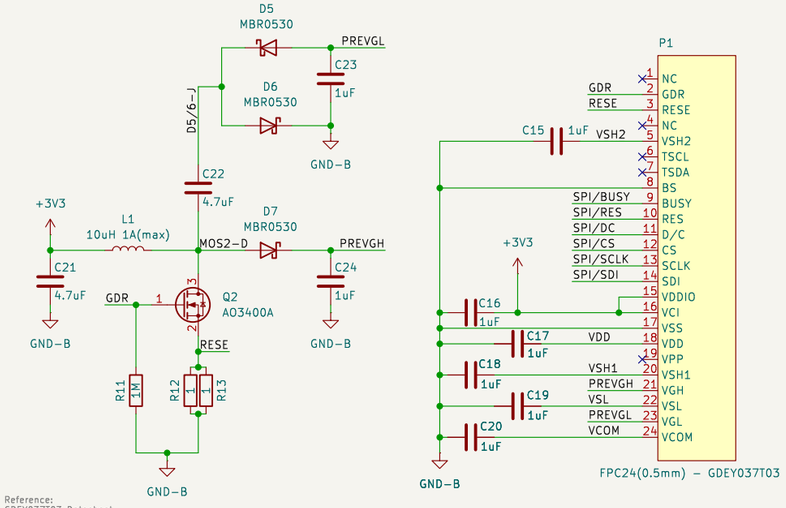
Reference:
How to Interface a Microcontroller with a Relay Using a MOSFET

Source: https://forum.digikey.com/t/how-to-interface-a-microcontroller-with-a-relay-using-a-mosfet/41470

Design reference on titles:
– MOSFET suitability
– Review of the MOSFET datasheet
– Schematic

PCB–A

EP DISPLAY

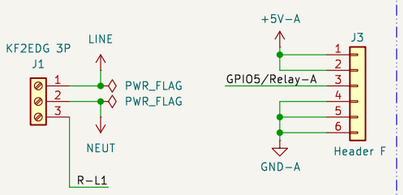


Reference:
GDEY037T03 Datasheet

Design reference on title:
10. Reference Circuit (pp.34)

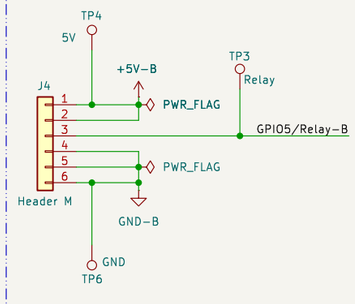
PCB–B

CONNECTORS–A



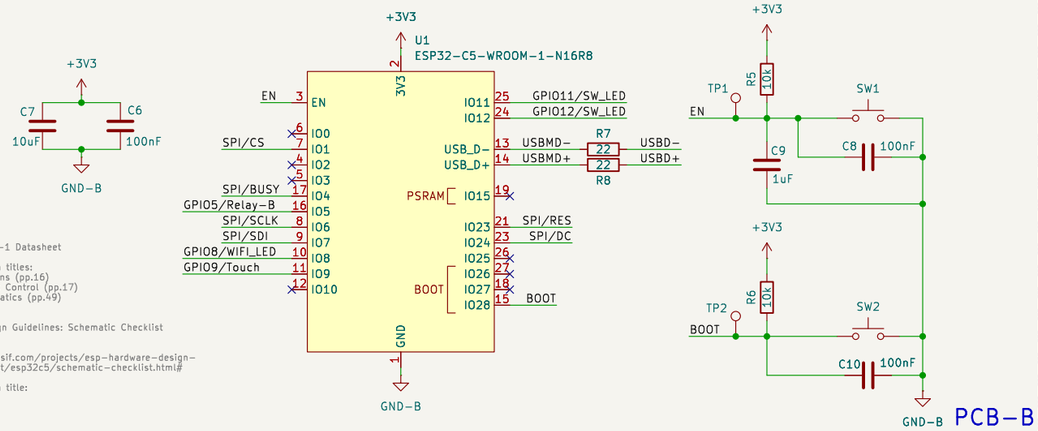
PCB–A

CONNECTORS–B



PCB–B

MCU



Reference:
ESP32–C5–WROOM–1 Datasheet

Design reference on titles:
4 Boot Configurations (pp.16)
4.1 Chip Boot Mode Control (pp.17)
9 Peripheral Schematics (pp.49)

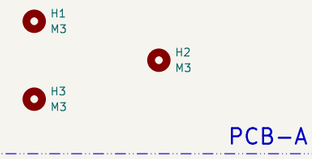
Reference:
ESP Hardware Design Guidelines: Schematic Checklist

Source:
https://docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/esp32c5/schematic-checklist.html#

Design reference on title:
– Strapping Pins
– USB

PCB–B

DRILL–A

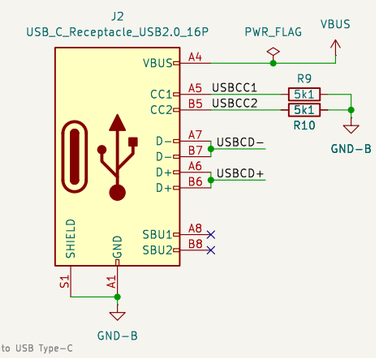


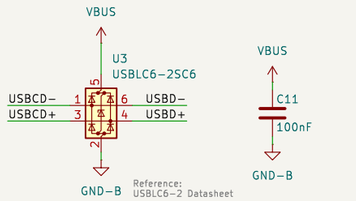
PCB–A

DRILL–B

PCB–B

USB–C CONNECTOR





Reference:
Introduction to USB Type–C

Source:
https://cdn.sparkfun.com/assets/e/b/4/f/7/USB-C_Datasheet.pdf

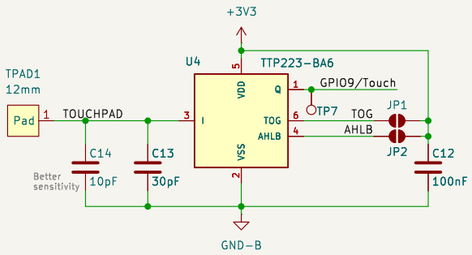
Design reference on title:
3.2 UFP Rd Pull–Down Resistors (pp.10)

Reference:
USBLC6–2 Datasheet

Design reference on title:
USBLC6–2 PCB layout considerations (pp.11)

PCB–B

TOUCH SENSOR



Reference:
Capacitive Touch Sensor Design Guide

Source:
https://ww1.microchip.com/downloads/aemDocuments/documents/TXFG/ApplicationNotes/ApplicationNotes/Capacitive-Touch-Sensor-Design-Guide-DS00002934-B.pdf

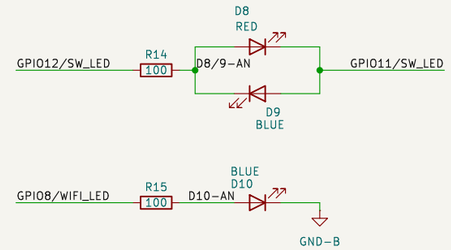
Design reference on title:
– Electrode Shapes (pp.5)
– Coplanar Ground Shield (pp.16)

Reference:
TTP223–BA6 Datasheet

Design reference on title:
APPLICATION CIRCUIT (pp.6)

PCB–B

LED INDICATORS



PCB–B

BrozuTechworks

Sheet: /
File: ESP32C5_iot_switch.kicad_sch

Title: ESP32–C5 Smart Switch

Size: A3	Date: 2026–03–15	**Rev: A0.1**
KiCad E.D.A. 9.0.7		Id: 1/1